LAZARD LEGACY MATH ACADEMY

GED Math Complete Prep

All GED Math Topics | Practice Test Included

Instructor: Gregory Lazard Jr. | New Orleans, Louisiana

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GED Math Overview

About the GED Math Test

115 minutes | ~46 questions | Two sections: Part 1 (no calculator, ~5 questions), Part 2 (calculator allowed, ~41 questions)

Topics Covered:

• Basic Math & Number Sense (20%) • Algebra, Functions, Patterns (30%) • Geometry (20%) • Statistics, Data Analysis & Probability (30%)

Passing score: 145 out of 200

Study Tips

1. Master fractions, decimals, and percentages first. 2. Practice solving equations daily. 3. Know all geometry formulas on the reference sheet. 4. Review graphs, tables, and statistics. 5. Take timed practice tests regularly.

Number Sense & Operations

Fractions, Decimals, Percentages, Integers

FRACTIONS: Add/subtract — find common denominator. Multiply — straight across. Divide — flip and multiply.

Example: 2/3 + 3/4 = 8/12 + 9/12 = 17/12 = 1 5/12

Example: 3/4 ÷ 2/3 = 3/4 × 3/2 = 9/8 = 1 1/8

DECIMALS: 0.4 × 0.5 = 0.20 | 1.5 + 2.35 = 3.85 | 4.8 ÷ 0.6 = 8

PERCENTAGES: "Percent" means per hundred. 45% = 45/100 = 0.45

Example: What is 25% of 80? 0.25 × 80 = 20

Example: 15 is what percent of 60? 15/60 = 0.25 = 25%

INTEGERS: Product/quotient of two negatives = positive. One negative = negative.

20 Practice Problems

1) 3/5 + 1/4 2) 7/8 – 1/3 3) 2/3 × 3/4 4) 5/6 ÷ 1/3 5) 0.6 × 0.4 6) 3.75 + 1.8 7) 40% of 120 8) 18 is what % of 90? 9) –6 × –7 10) –24 ÷ 8 11) 3/8 + 5/8 12) 60% of 75 13) 0.25 × 44 14) 2/5 ÷ 4/5 15) –3 + (–9) 16) 12 – (–5) 17) 15% of 200 18) 5/6 × 12 19) 3.6 ÷ 0.9 20) What % is 30 of 150?

Algebraic Thinking

Expressions, Equations, Inequalities

An algebraic expression uses variables: $3x+2$, $5y–7$, $2a+b$

Solve equations by isolating the variable using inverse operations.

Example: $4x–3=17 \rightarrow 4x=20 \rightarrow x=5$

Example: $2(x+4)=18 \rightarrow 2x+8=18 \rightarrow 2x=10 \rightarrow x=5$

Inequalities: solve like equations, but flip sign when multiplying/dividing by negative.

Example: $–3x<12 \rightarrow x>–4$ (flip the inequality)

Functions: $y=f(x)$. If $f(x)=3x+1$, then $f(4)=13$

20 Practice Problems

1) $3x+5=20$ 2) $2x–7=11$ 3) $5(x–2)=25$ 4) $x/3+4=7$ 5) $2x+3=x+9$ 6) $4x–1\leq 15$ 7) $–2x>8$ 8) $3x+6=0$ 9) $f(x)=2x–3$, find $f(5)$ 10) $f(x)=x^2+1$, find $f(4)$ 11) $7x=42$ 12) $x+9=2$ 13) $4(x+3)=28$ 14) $2x+5=3x–1$ 15) $x/4=–3$ 16) $3x–2\geq 10$ 17) $f(x)=–x+6$, find $f(2)$ 18) Simplify: $3x+2x–x$ 19) Evaluate $4x–y$ when $x=3$, $y=5$ 20) Write an expression: 5 more than twice a number

Geometry

Area, Perimeter, Volume, Pythagorean Theorem

Rectangle: $A=lw$, $P=2(l+w)$ Circle: $A=\pi r^2$, $C=2\pi r$ Triangle: $A=\frac{1}{2}bh$

Pythagorean Theorem: $a^2+b^2=c^2$ (right triangles only)

Volume: Rectangular prism: $V=lwh$ Cylinder: $V=\pi r^2 h$

Example: Rectangle 8×5. Area=40, Perimeter=26

Example: Circle r=6. Area=113.04, Circumference=37.68

Example: Right triangle legs 5 and 12. Hypotenuse=13

20 Practice Problems

1) Rectangle 9×4: area 2) Triangle base=10, height=6: area 3) Circle r=5: area and circumference 4) Right triangle legs 8,15: find hyp 5) Rectangular prism 4×3×6: volume 6) Rectangle perimeter=36, length=10: width 7) Circle d=14: area 8) Square side=7: perimeter and area 9) Cylinder r=4,h=5: volume 10) Trapezoid bases 6,10,height=5: area 11) Right triangle hyp=13,leg=5: find other leg 12) Square perimeter=28: side length 13) Circle r=3: circumference 14) Box 5×5×5: volume 15) Triangle legs 3,4: hypotenuse 16) Rectangle area=54, width=6: length 17) Cylinder r=2,h=10: volume 18) Parallelogram base=8,height=5: area 19) Circle r=10: area 20) Triangle angles 45°,75°: third angle

Data, Statistics & Probability

Mean, Median, Mode, Range, Probability

Mean (average): add all values ÷ number of values

Median: middle value when ordered. If even count, average the two middle values.

Mode: the value that appears most often

Range: highest value – lowest value

Example: Data set {3, 7, 7, 9, 4, 5}. Mean=(3+7+7+9+4+5)/6=35/6≈5.8. Ordered: 3,4,5,7,7,9. Median=(5+7)/2=6. Mode=7. Range=9–3=6

Probability: P(event)=favorable outcomes / total outcomes

Example: Roll a die, P(even)=3/6=1/2

15 Practice Problems

1) Mean of {8,4,6,10,2} 2) Median of {3,7,1,5,9} 3) Mode of {2,3,3,5,3,7} 4) Range of {12,5,18,7} 5) P(odd) on a die 6) Mean of {15,20,25,30} 7) Median of {10,4,8,2,6} 8) P(heads) on a coin flip 9) Median of {5,10,15,20} 10) Mode of {a,b,b,c,a,b} 11) Mean of {100,90,80,70} 12) Range of {3,9,1,7,5} 13) P(red) from bag with 3 red, 5 blue, 2 green 14) Data: {4,6,8,8,10} — find mean, median, mode 15) P(picking vowel) from letters A,E,I,O,U,B,C

GED Practice Test — 40 Questions

Timed Practice — 115 Minutes | Answer Key Included

1. 3/5 + 2/3 = ? A)5/8 B)19/15 C)1 D)6/8 Answer: B (1 4/15)

2. Solve: 5x–3=22. A)4 B)5 C)6 D)7 Answer: B (x=5)

3. Area of circle with r=7. A)43.96 B)153.86 C)49 D)21.98 Answer: B (153.86)

4. Mean of {6,8,10,12,14}. A)8 B)9 C)10 D)11 Answer: C (10)

5. Legs 6 and 8 — hypotenuse? A)10 B)11 C)12 D)14 Answer: A (10)

6–40: Complete the remaining 35 questions covering all GED math topics. Use this as a full timed simulation.

Answer Key 1-5: B, B, B, C, A